ARYAN SINGH

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SUMMARY

Computer Science and Mathematics student at UIUC with a strong foundation in data structures, algorithms, and Linux/UNIX systems, and hands-on experience building scalable, distributed, and parallel software in C, C++, Python, and CUDA. Proven ability to design, automate, and operate production-grade infrastructure end to end — from an FPGA job-scheduling accelerator to a from-scratch GPU inference engine — alongside a 1st-place finish (out of 400 students) in a memory-optimization contest. Eager to apply rigorous engineering and strong computer science fundamentals to large-scale, real-world software challenges.

EDUCATION

University of Illinois Urbana-Champaign — Urbana, IL

B.S., Computer Science and Mathematics | GPA: 3.78 / 4.00 — Expected Graduation May 2028

- Relevant coursework: Data Structures & Algorithms, Distributed Systems, Parallel Programming (CUDA), Computer Architecture, Systems Programming, Machine Learning, Artificial Intelligence, Database Systems
- Honors: 1st place out of 400 students, CS 341 Malloc Contest (highest-performing memory allocator); 3rd place, UIUC Capture the Flag (CTF) Competition 2025; 5th place out of 50 schools, Robot Tour, UIUC State Science Olympiad 2024

EXPERIENCE

University of Illinois Chicago — Chicago, IL

Software Engineering Intern — May 2025 – Aug 2025

- Engineered the data-movement layer for a job-scheduling accelerator on a Xilinx Alveo U55C FPGA. Implemented PCIe/DMA transfer paths and on-device memory architecture (C++, Vitis HLS) that route jobs into and out of the compute kernel.
- Drove the shift from batch execution to a continuous data streaming pipeline alongside the algorithm's research scientists, eliminating per-batch startup and teardown overhead and sustaining 95%+ FPGA utilization.
- Built an asynchronous, multithreaded C++ host runtime that overlaps job ingestion with in-flight execution, removing the host-side stalls that starve accelerators in naive designs.
- Owned validation end to end: learned the Xilinx/Vitis ecosystem independently, built integration tests against real hardware, and presented the completed system to research stakeholders.

PROJECTS

Advanced Systems Programming (CS 341) | C, Linux/UNIX, POSIX Threads, TCP/IP

- Memory Allocator (malloc): Architected a custom dynamic memory allocator (malloc, free, realloc, calloc) in C, optimizing block splitting, coalescing, and caching to win 1st place out of 400 students for peak memory throughput and minimal fragmentation.
- Network Server (nonstop networking): Built an asynchronous, multi-client Linux server using epoll and raw TCP/IP sockets to handle concurrent data streams, deeply exploring network state machines and protocol edge cases.
- UNIX Shell (shell): Developed a fully functional POSIX shell in C from scratch, implementing process creation (fork, exec), signal handling, I/O redirection, and advanced shell commands.

- Parallel Execution (parallel make & mapreduce): Engineered a multi-threaded make clone and a local MapReduce framework, utilizing mutexes and condition variables to resolve complex concurrency issues and prevent deadlocks.

GPT-2 Inference Engine from Scratch | CUDA C++ gpt2.aryan-singh.dev

- Architected a complete GPT-2 inference engine in pure CUDA C++ with no ML frameworks; every transformer stage (attention, matmul, layer norm, sampling) is a custom GPU kernel.
- Drove generation throughput to **350 tokens/second** on NVIDIA A40 GPUs, a **24.2x** end-to-end speedup and **117x** peak decode speedup over the naive baseline.
- Cut per-token latency with KV caching, FlashAttention-2-style fused attention, WMMA tensor cores, and CUTLASS fused epilogues, each selected via Nsight profiling and verified for zero accuracy loss.

Homelab: Self-Hosted Production Infrastructure | Proxmox, Kubernetes (K3s), OPNsense, Cloudflare

- Built and operate a self-hosted platform running **25+** services across **50+** containers on a Proxmox virtualization layer and K3s cluster, from personal cloud storage to the public project demos linked on this resume.
- Exposes those services to the public internet safely through layered security: an OPNsense firewall with Suricata and CrowdSec intrusion detection, Cloudflare mTLS at the edge, and Authelia single sign-on in front of every app.
- Maintains full observability with Grafana, Loki, and Prometheus, including GeoIP-enriched dashboards that surface and help block malicious traffic hitting the firewall and reverse proxy in real time.
- Ships every personal project through Jenkins CI/CD integrated with the GitHub Checks API: each push builds, tests, and deploys to the cluster automatically.

Sentinel: AI Infrastructure Monitoring Agent | Python, MCP, Prometheus, Grafana/Loki sentinel.aryan-singh.dev

- Built an MCP-based agent that monitors the homelab's services and infrastructure, pulling live metrics and logs from Prometheus, Loki, and Grafana to detect anomalies and diagnose issues in natural language.
- Exposed infrastructure operations as MCP tools — service health checks, container inspection, and log queries across the Proxmox and K3s stack — so the agent can investigate incidents and surface root-cause explanations on demand.
- Turns noisy alerts into concise, actionable summaries by correlating signals across services to pinpoint the failing component, rather than paging on raw thresholds.

Vega: Multi-Agent Autonomous Trading System | Python, LangGraph, MLX, Alpaca API GitHub

- Pushed a locally hosted 20B-parameter model far beyond its size class with tools and agentic scaffolding: it researches markets overnight and paper-trades at the open, no human in the loop.
- Built custom MCP servers for web search, live market data (Finnhub), and a Python execution environment, so the model pulls real-time data and writes and runs its own trading algorithms.
- Learns from its own record: an AI committee debates every position, a bias-aware workflow vets internet sources, and post-trade analysis of losing picks feeds back into each strategy.

TECHNICAL SKILLS

Languages: C, C++, Python, CUDA, Rust, Java, TypeScript, SQL

AI/ML and Data: LangChain, RAG, Apple MLX, NumPy, Pandas, Model Context Protocol (MCP)

Backend and Systems: Node.js, FastAPI, Spring Boot, PostgreSQL, pgvector, Redis, Linux

Infrastructure and DevOps: Docker, Kubernetes (K3s), Proxmox, Jenkins CI/CD, Grafana/Loki, Prometheus

CERTIFICATIONS

Anthropic Academy: AI Fluency (Framework and Foundations); Introduction to Model Context Protocol (MCP)

Hugging Face: AI Agents Course; LLM Course